

Algorithms 2

Lovasz Local Lemma

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1. Lovasz Local Lemma

Definition 1 The scenario:

Let $\mathcal{E}_1, \dots, \mathcal{E}_n$ be bad events (over the same sample space).

We wish to prove: $\mathbb{P}\left[\bigcap_{i=1}^n \overline{\mathcal{E}_i}\right] > 0$ – no bad event occurs.

- If $\mathcal{E}_1, \dots, \mathcal{E}_n$ are pairwise independent, then we can argue as follows:
 - Recall: events A and B are independent if $\mathbb{P}[A \cap B] = \mathbb{P}[A]\mathbb{P}[B]$.
 - If A, B are independent, then so are $\overline{A}, \overline{B}$:

$$\begin{aligned}\mathbb{P}[\overline{A} \cap \overline{B}] &= 1 - \mathbb{P}[A \cup B] \\ &= 1 - \mathbb{P}[A] - \mathbb{P}[B] + \mathbb{P}[A \cap B] \\ &= (1 - \mathbb{P}[A])(1 - \mathbb{P}[B]) \\ &= \mathbb{P}[\overline{A}] \mathbb{P}[\overline{B}]\end{aligned}$$

1.2 Pairwise independence is not enough

- Back to our problem:
- $\mathcal{E}_1, \dots, \mathcal{E}_n$ pairwise independent $\Rightarrow \overline{\mathcal{E}}_1, \dots, \overline{\mathcal{E}}_n$ pairwise independent.
- If $\mathbb{P}[\mathcal{E}_i] < 1$ for all $i \in [n]$, then:

$$\mathbb{P}\left[\bigcap_{i=1}^n \overline{\mathcal{E}}_i\right] = \prod_{i=1}^n \mathbb{P}[\overline{\mathcal{E}}_i] = \prod_{i=1}^n (1 - \mathbb{P}[\mathcal{E}_i]) > 0$$

(assuming this equality holds).

- But pairwise independence does **not** imply

$$\mathbb{P}\left[\bigcap_{i=1}^n \overline{\mathcal{E}}_i\right] = \prod_{i=1}^n \mathbb{P}[\overline{\mathcal{E}}_i].$$

- Mutual independence does.
- What if $\mathcal{E}_1, \dots, \mathcal{E}_n$ do depend on one another in some way?
- The Lovasz local lemma provides a partial solution.

An experiment is made, by throwing 2 fair coins independently of each other.

- C_1 the event that the first coins is heads.
- C_2 the event that the second coint is heads.
- C_3 the event that the outcome of the coins is not the same.

Clearly C_1 and C_2 are independent. Also C_3 is independent of both C_1 and C_2

$$\begin{aligned}\mathbb{P}(C_1 \wedge C_3) &= \mathbb{P}(C_3 \mid C_1) \cdot \mathbb{P}(C_1) \\ &= \frac{1}{2} \cdot \frac{1}{2} \\ &= \mathbb{P}(C_1) \cdot \mathbb{P}(C_3)\end{aligned}$$

But

$$\mathbb{P}(C_1 \wedge C_2 \wedge C_3) = 0 \neq \frac{1}{8} = \mathbb{P}(C_1) \cdot \mathbb{P}(C_1) \cdot \mathbb{P}(C_1)$$

Definition 2 Events $\mathcal{E}_1, \dots, \mathcal{E}_n$ are said to be **mutually independent** if

$$\mathbb{P} \left[\bigcap_{i \in I} \mathcal{E}_i \right] = \prod_{i \in I} \mathbb{P}[\mathcal{E}_i]$$

whenever $\emptyset \neq I \subseteq [n]$.

Definition 3 (Alternative definition) Event A is mutually independent of events $\mathcal{E}_1, \dots, \mathcal{E}_n$ if

$$\mathbb{P} \left[A \mid \bigcap_{i=1}^n B_i \right] = \mathbb{P}[A]$$

where $B_i \in \{\mathcal{E}_i, \overline{\mathcal{E}_i}\}$ for all $i \in [n]$.

- No matter whether each $\mathcal{E}_i$ occurs or not, this does not affect $\mathbb{P}[A]$.
- For a proof of equivalence, see the booklet.

- Mutual independence $\Rightarrow$ pairwise independence.
- The converse implication fails.

Example. Let $\Omega := \{1, 2, 3, 4\}$ with the discrete uniform measure. $A := \{1, 2\}$, $B := \{1, 3\}$, $C := \{2, 3\}$.

- $\mathbb{P}[A] = \mathbb{P}[B] = \mathbb{P}[C] = \frac{1}{2}$.
- A, B are independent: $\mathbb{P}[A \cap B] = \mathbb{P}[\{1\}] = \frac{1}{4} = \mathbb{P}[A] \cdot \mathbb{P}[B]$.
- The same applies to B, C and A, C .
- Yet $\mathbb{P}[A \cap B \cap C] = \mathbb{P}[\emptyset] = 0 \neq \mathbb{P}[A] \cdot \mathbb{P}[B] \cdot \mathbb{P}[C]$.
- So A, B, C are pairwise independent but **not** mutually independent.

- Mutual independence is a **stronger** form of independence than pairwise independence.
- We have seen that $\mathcal{E}_1, \dots, \mathcal{E}_n$ pairwise independent $\Rightarrow \overline{\mathcal{E}_1}, \dots, \overline{\mathcal{E}_n}$ pairwise independent.
- This property is also supported in mutual independence:

Theorem 1 If $\mathcal{E}_1, \dots, \mathcal{E}_n$ are mutually independent, then so are $\overline{\mathcal{E}_1}, \dots, \overline{\mathcal{E}_n}$.

For a proof, see the booklet.

- The following is a handy tool for determining mutual independence.

Theorem 2 (Mutual independence principal)

Let $X := (X_1, \dots, X_m)$ be a sequence of pairwise independent trials. Let $A_1, \dots, A_n$ be events, and for each $i \in [n]$ let A_i be determined by trials $F_i \subseteq X$. Given $I \subseteq [n]$ and $j \in [n] \setminus I$, if

$$F_j \cap \left(\bigcup_{i \in I} F_i \right) = \emptyset,$$

then A_j is mutually independent of $(A_i)_{i \in I}$.

- There are several versions. We state one of those.
- The version stated here is called the **symmetric version**.
- We focus on applications of the lemma and not its proof(s).

The dependency graph

Definition 4 Events $\mathcal{E}_1, \dots, \mathcal{E}_n$. A dependency graph $D := D(\mathcal{E}_1, \dots, \mathcal{E}_n)$ is any graph whose vertex set is $\mathcal{E}_1, \dots, \mathcal{E}_n$ and supporting the property: for every $i \in [n]$, event $\mathcal{E}_i$ is mutually independent of the events $\{\mathcal{E}_j : j \neq i \wedge \{i, j\} \notin E(D)\}$.

Remark

Sometimes a directed version of such graphs is more useful (i.e., in **asymmetric** applications).

Remark

Danger: We did **not** just say that we build a graph on $\mathcal{E}_1, \dots, \mathcal{E}_n$ by placing an edge $(\mathcal{E}_i, \mathcal{E}_j)$ if $\mathcal{E}_i, \mathcal{E}_j$ are dependent.

Theorem 3 (The symmetric local lemma)

Let $\mathcal{E}_1, \dots, \mathcal{E}_n$ be events such that the following holds:

1. **(Symmetry)** $\mathbb{P}[\mathcal{E}_i] \leq p$ for all $i \in [n]$.
2. **(Limited dependency)** $\Delta(D(\mathcal{E}_1, \dots, \mathcal{E}_n)) \leq d$.
3. **(Bound p and d)** $e \cdot p \cdot (d + 1) \leq 1$ (up to a constant).

Then $\mathbb{P}\left[\bigcap_{i=1}^n \overline{\mathcal{E}_i}\right] > 0$.